Project: Worked examples
Title: Duct sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates duct sizes by the equal-friction or the velocity method, rounded up to standard sizes. It reports the velocity, friction rate and losses of each section and the pressure drop along the index run.

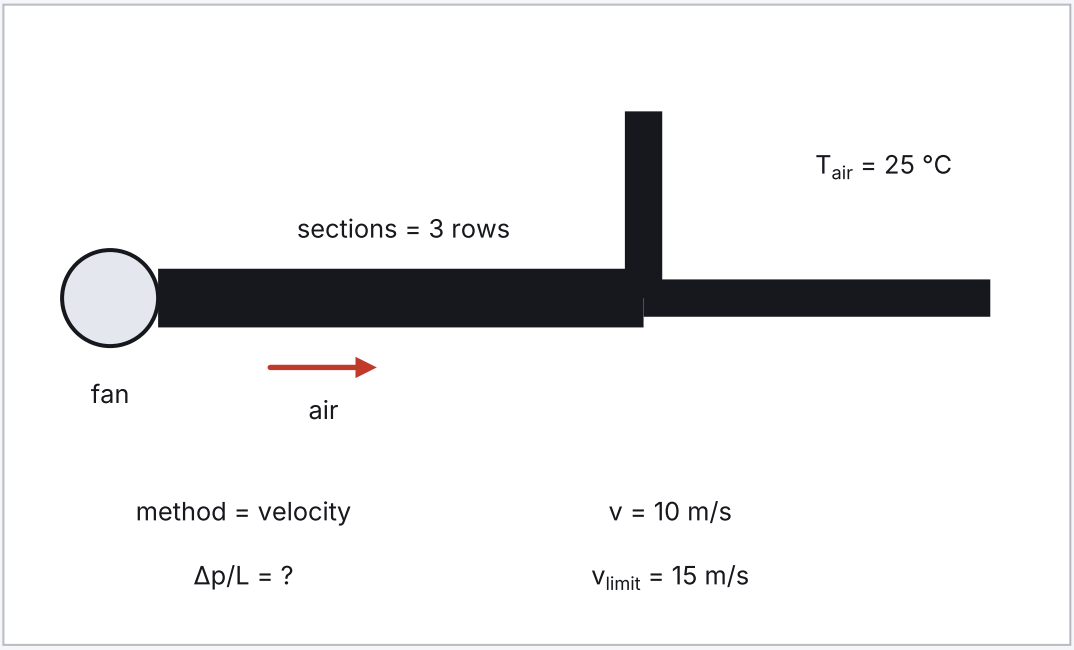


Figure 1 — A duct run from the fan through its sections as listed (a main and two branches drawn), each sized by the chosen method. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Round duct diameter from the design target	D by Colebrook with the air properties at T (equal friction), or D = √(4Q/πv) (velocity), rounded up to the next standard size
Rectangular duct of equal friction and capacity	D_e = 1.30 (ab)^0.625/(a + b)^0.25
Flat-oval duct of equal friction and capacity	D_e = 1.55 A^0.625/P^0.25, with A = πb²/4 + b(a − b) and P = πb + 2(a − b)
Velocity method, rectangular or flat-oval duct with side a fixed	b from the free area A = Q/v (rectangle b = A/a; flat oval the smaller root of (1 − π/4)b² − ab + A = 0), rounded up; D_e is kept for the friction only
Friction loss per metre, at the velocity v_e of the equivalent round duct	$\Delta p/L = f \rho v_e^2/(2D_e)$
Fitting losses, at the actual velocity	$\Delta p = \Sigma C \rho v^2/2$
Index-run totals, summed over every section of the run	$\Delta p_{run} = \Sigma(\Delta p/L \cdot L + \Sigma C \rho v^2/2)$

Basis: Duct design (friction chart basis, equivalent diameters, loss coefficients)[1](Ch. 21); Flow of fluids in pipes and ducts[2](§4); Ventilation for buildings: sheet metal air ducts and fittings with rectangular cross-section, dimensions[3]; Ventilation for buildings: sheet metal air ducts and fittings with circular cross-section, dimensions[4].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Sizing method	velocity		assumption: Velocity method for an extract system.	any
sections	Duct sections	3 row(s)		drawing: DRG-M-310 rev A	any
$round_{to}$	Round sizes up to a multiple of	50	mm	spec: BS EN 1506 preferred sizes	1 … 100 mm
eps_{rough}	Duct roughness	0.15	mm	assumption: Spiral-wound galvanised duct with joints.	0.03 … 3 mm
T_{air}	Air temperature	25	°C	assumption: Workshop air 25 °C.	-20 … 60 °C
p_{atm}	Barometric pressure	100.5	kPa	assumption: Site 70 m AOD, ≈ 100.5 kPa.	80 … 105 kPa
dp_{target}	Target friction rate (equal friction)	1	Pa/m	default	0.3 … 5 Pa/m
v_{target}	Target velocity (velocity method)	10	m/s	spec: Dust-carrying extract: 10 m/s minimum transport velocity, SPEC-M-09	2 … 20 m/s
v_{limit}	Velocity limit (warning and check)	15	m/s	spec: SPEC-M-09 noise limit	2 … 30 m/s
dp_{limit}	Total pressure-drop limit (check)	600	Pa	spec: Extract fan external static 600 Pa	50 … 2000 Pa

TABLE ROW	VALUES
sections[1]	name = hood branch; Q = 0.35 m³/s; L = 6 m; shape = round; a = —; D = —; C = 1.5; dp_eq = 120 Pa; index = yes
sections[2]	name = main; Q = 1.05 m³/s; L = 20 m; shape = round; a = —; D = —; C = 0.6; dp_eq = 0 Pa; index = yes
sections[3]	name = stack; Q = 1.05 m³/s; L = 8 m; shape = round; a = —; D = —; C = 1; dp_eq = 30 Pa; index = yes

3. Assumptions

A1. Air is dry at the stated temperature and pressure. (Ideal gas, ρ = p/(287.055 T).)
A2. The viscosity of air follows Sutherland's law. (Standard air properties.)
A3. Rectangular and flat-oval ducts are sized through the equivalent diameter for equal friction and capacity. (Huebscher / Heyt–Diaz relations as given in ASHRAE Fundamentals Ch. 21.)
A4. Fitting loss coefficients C are user-supplied (manufacturer or ASHRAE/CIBSE data, not embedded) and refer to the section's velocity pressure. (Non-negotiable: no standard tables embedded.)

A5. method (Sizing method) is taken as velocity. (Velocity method for an extract system.)
A6. eps_{rough} (Duct roughness) is taken as 0.15 mm. (Spiral-wound galvanised duct with joints.)
A7. T_{air} (Air temperature) is taken as 25 °C. (Workshop air 25 °C.)
A8. p_{atm} (Barometric pressure) is taken as 100.5 kPa. (Site 70 m AOD, ≈ 100.5 kPa.)

4. Working

[10] ρ = p / (287.055 T)
= 100500 Pa / (287.055 × 298.15 K)
= 1.1743 kg/m³
Air density

[11] μ = Sutherland's law
= T = 298.15 K
= 18.371 uPa·s
Air dynamic viscosity

[12] D_{s1} = hood branch: diameter
= target 211.1 mm rounded up to 50 mm
= 250 mm
Diameter, hood branch

[13] v_{s1} = hood branch: Q / A
= 0.35 / 0.049087
= 7.1301 m/s
Velocity, hood branch

[14] Re_{s1} = hood branch: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.1743 × 7.1301 × 0.25 / 0.000018371
= 113940
Reynolds number, hood branch

[15] f_{s1} = hood branch: friction factor
= Colebrook–White
= 0.02041
Friction factor, hood branch

[16] dpL_{s1} = hood branch: f ρ v_e^2 / (2 D_e)
= 0.02041 × 1.1743 × 7.1301² / (2 × 0.25)
= 2.4369 Pa/m
Friction rate, hood branch

[17] $dp_{f,s1}$ = hood branch: (Δp/L) L
= 2.4369 × 6
= 14.622 Pa
Friction loss, hood branch

[18] $dp_{fit,s1}$ = hood branch: ΣC ρ v²/2
= 1.5 × 29.849
= 44.774 Pa
Fitting loss, hood branch

[19] dp_{s1} = hood branch: friction + fittings + equipment
= 14.622 + 44.774 + 120
= 179.4 Pa
Section loss, hood branch (index run)

[20] D_{s2} = main: diameter
= target 365.64 mm rounded up to 50 mm
= 400 mm
Diameter, main

[21] v_{s2} = main: Q / A
= 1.05 / 0.12566
= 8.3556 m/s
Velocity, main

[22] Re_{s2} = main: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.1743 × 8.3556 × 0.4 / 0.000018371
= 213630
Reynolds number, main
[23] f_{s2} = main: friction factor
= Colebrook–White
= 0.018028
Friction factor, main
[24] dpL_{s2} = main: f ρ v_e^2 / (2 D_e)
= 0.018028 × 1.1743 × 8.3556² / (2 × 0.4)
= 1.8475 Pa/m
Friction rate, main
[25] $dp_{f,s2}$ = main: (Δp/L) L
= 1.8475 × 20
= 36.95 Pa
Friction loss, main
[26] $dp_{fit,s2}$ = main: ΣC ρ v²/2
= 0.6 × 40.992
= 24.595 Pa
Fitting loss, main
[27] dp_{s2} = main: friction + fittings + equipment
= 36.95 + 24.595 + 0
= 61.545 Pa
Section loss, main (index run)
[28] D_{s3} = stack: diameter
= target 365.64 mm rounded up to 50 mm
= 400 mm
Diameter, stack
[29] v_{s3} = stack: Q / A
= 1.05 / 0.12566
= 8.3556 m/s
Velocity, stack
[30] Re_{s3} = stack: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.1743 × 8.3556 × 0.4 / 0.000018371
= 213630
Reynolds number, stack
[31] f_{s3} = stack: friction factor
= Colebrook–White
= 0.018028
Friction factor, stack
[32] dpL_{s3} = stack: f ρ v_e^2 / (2 D_e)
= 0.018028 × 1.1743 × 8.3556² / (2 × 0.4)
= 1.8475 Pa/m
Friction rate, stack
[33] $dp_{f,s3}$ = stack: (Δp/L) L
= 1.8475 × 8
= 14.78 Pa
Friction loss, stack
[34] $dp_{fit,s3}$ = stack: ΣC ρ v²/2
= 1 × 40.992
= 40.992 Pa
Fitting loss, stack

[35] dp_{s3} = stack: friction + fittings + equipment
= 14.78 + 40.992 + 30
= 85.772 Pa
Section loss, stack (index run)
[36] dp_{index} = Σ section losses on the index run
= 3 section(s)
= 326.71 Pa
Index-run pressure drop
[37] dp_{all} = Σ section losses, all sections
= 3 section(s)
= 326.71 Pa
Sum over every section (not a path)
[38] v_{peak} = highest section velocity
=
= 8.3556 m/s
Highest velocity
[39] u_{dp} = dp_{index} / dp_{limit}
= 326.71 / 600
= 0.54452
Pressure-drop utilisation
[40] u_v = v_{peak} / v_{limit}
= 8.3556 / 15
= 0.55704
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
dp_{index}	326.7 Pa	Index-run Δp
v_{peak}	8.356 m/s	Highest velocity (governing)
D_{s1}	250 mm	D, hood branch
v_{s1}	7.130 m/s	v, hood branch
Re_{s1}	113940	Re, hood branch
f_{s1}	0.02041	f, hood branch
dpL_{s1}	2.437 Pa/m	Δp/L, hood branch
$dp_{f,s1}$	14.62 Pa	Friction loss, hood branch
$dp_{fit,s1}$	44.77 Pa	Fitting loss, hood branch
dp_{s1}	179.4 Pa	Section loss, hood branch
D_{s2}	400 mm	D, main
v_{s2}	8.356 m/s	v, main
Re_{s2}	213630	Re, main
f_{s2}	0.018028	f, main
dpL_{s2}	1.848 Pa/m	Δp/L, main
$dp_{f,s2}$	36.95 Pa	Friction loss, main
$dp_{fit,s2}$	24.59 Pa	Fitting loss, main
dp_{s2}	61.55 Pa	Section loss, main

SYMBOL	VALUE	DESCRIPTION
D_{s3}	400 mm	D, stack
v_{s3}	8.356 m/s	v, stack
Re_{s3}	213630	Re, stack
f_{s3}	0.018028	f, stack
dpL_{s3}	1.848 Pa/m	Δp/L, stack
$dp_{f,s3}$	14.78 Pa	Friction loss, stack
$dp_{fit,s3}$	40.99 Pa	Fitting loss, stack
dp_{s3}	85.77 Pa	Section loss, stack
dp_{all}	326.7 Pa	Σ all sections

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Highest velocity against limit	8.356 m/s	15 m/s	0.56	
PASS	Index-run pressure drop against limit	326.7 Pa	600 Pa	0.54	

7. Applicability

None.

Defaults used (inputs left empty):
dp_{target}: not entered, so the default of 1 Pa/m is used.

8. Verification

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Template bs.duct-sizing@1.2.1 Content hash a0afb84d934b74eb Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
3. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Rectangular Cross-Section—Dimensions. BS EN 1505. BSI.
4. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Circular Cross-Section—Dimensions. BS EN 1506. BSI.